MIL-C-81461E(AS)
23 February 1984
SUPERSEDING
MIL-C-81461(AS)
15 April 1967
(See 6.11)

MILITARY SPECIFICATON

COMPENSATOR GROUP, MAGNETIC AN/ASA-65(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers the AN/ASA-65(V) semi-automatic magnetic compensation system herein referred to as the equipment, which is used in aircraft equipped with Magnetic Airborne Detection (MAD) equipment.

1.2 Classification - The equipment covered by this specification includes five installation configurations consisting of the following items:

a. COMPENSATOR GROUP, MAGNETIC AN/ASA-65(V)1

Unit	Type Designation	Applic. Para.	Qty.
Control-Indicator	C-7718/ASA-65(V)	3.5.1	1
Amplifier, Electronic Control	AM-6056/ASA-65(V)	3.5.2	1
Magnetometer Assy.	DT-355/ASA-65(V)	3.5.2	1
Coil Assembly	MX-8130/ASA-65(V)	3.5.4	1

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: The Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD), Code 93, Lakehurst, NJ 08733 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document or by letter.

FSC 5845

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

b. COMPENSATOR GROUP, MAGNETIC AN/ASA-65(V)2

Unit	Type Designation	Applic. Para.	Qty.
Control-Indicator	C-8935/ASA-65(V)	3.5.1	1
Amplifier, Electronic	AM-6459/ASA-65(V)	3.5.2	1
Magnetometer Assy.	DT-355/ASA-65(V)	3.5.5	3
Coil, Magnetic Compensating	MX-8897/ASA-65(V)		

c. COMPENSATOR GROUP, MAGNETIC AN/ASA-65(V)3 and AN/ASA-65A(V3)

Unit	Type Designation	Applic. Para.	Qty.
Control-Indicator or	C-8936/ASA-65(V) C-8936A/ASA-65(V)	3.5.1	1
Amplifier, Electronic Control	AM-6460/ASA-65(V)	3.5.2	1
Magnetometer Assy.	DT-355/ASA-65(V)	3.5.3	1
Coil, Magnetic Compensating, L	MX-9057/ASA-65(V)	3.5.6	1
Coil, Magnetic Compensating, T,V	MX-9058/ASA-65(V)	3.5.7	2

d. COMPENSATOR GROUP, MAGNETIC AN/ASA-65(V)4

Unit	Type Designation	Applic. Para.	Qty.
Control-Indicator or	C-8936/ASA-65(V) C-8936A/ASA-65(V)	3.5.1	1
Amplifier, Electronic Control	AM-6460/ASA-65(V)	3.5.2	1
Magnetometer Assy.	DT-355/ASA-65(V)	3.5.3	1
Coil, Magnetic Compensating, L	MX-10079/ASA-65(V)	3.5.8	1
Coil, Magnetic Compensating, T,V	MX-9058/ASA-65(V)	3.5.7	2
Computer Magnetic Field	CP-1390/ASA-65(V)	3.5.9	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

Indicator, Magnetic Field	ID-2205/ASA-65(V)	3.5.10	1
Mounting Base	MT-6086/ASA-65(V)	3.5.11	1

e. COMPENSATOR GROUP, MAGNETIC AN/ASA-65(V)5

Unit	Type Designation	Applic. Para.	Qty.
Control-Indicator	C-8935/ASA-65(V)	3.5.1	1
Amplifier, Electronic Control	AM-6459/ASA-65(V)	3.5.2	1
Magnetometer Assembly	DT-355/ASA-65(V)	3.5.3	1
Coil, Magnetic Compensation	MX-8897/ASA-65(V)	3.5.5	3
Computer, Magnetic Field	CP-1390/ASA-65(V)	3.5.9	1
Indicator, Magnetic Field	ID-2254/ASA-65(V)	3.5.10	1
Mounting Base	MT-6174/ASA-65(V)	3.5.12	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

1.3 Associated equipment - The AN/ASA-65(V) Magnetic Compensator Group should operate with the associated equipmented listed in 6.8.

2. APPLICABLE DOCUMENTS

2.1 Government Documents

2.1.1 Specifications and standards - Unless otherwise specificed, the following specifications and standards of the issue listed in that issue of the Department of Defence Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-5400	Electronic Equipment, Aerospace, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment, Rack or Console Mounted
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Accessories Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment including Ground Support Equipment
MIL-S-19500	Semiconductor Devices, General Specifications for
MIL-M-38510	Microcircuit, General Specification for
MIL-C-81511	Connector, Electric, Circular, High Density, Quick Disconnect, Environment Resisting, and Accessories, General Specification for

NAVAL AIR SYSTEMS COMMAND

WR-101	Part I - Electromagnetic Control Requirements for Advanced ASW Avionics System, A-New

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-595	Color

MILITARY

MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-461	Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristic, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-701	List of Standard Semiconductor Devices
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-831	Test Reports, Preparation of
MIL-STD-965	Parts Control Program
MIL-STD-2084	General Requirements for Maintainability of Avionic and Electronic System and Equipment

(Copies of specifications, standards, handbooks, drawings, and publications, required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 First article - When specified, a sample(s) shall be subjected to first article inspection. (See 4.3 and 6.2.3).

3.2 Parts and materials - In the selection of parts and materials fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. Microelectric items shall conform to the requirements specified herein.

b. Other parts and material requirements shall conform to MIL-E-5400.

c. Non-repairable subassemblies shall be used in accordance with MIL-STD-2084 and as specified in MIL-E-5400.

3.2.1 Non-standard part approval. Approval for the use of non-standard parts shall be in accordance with MIL-STD-965, Procedure I.

3.2.2 Electron devices. Transistors and diodes shall be chosen and applied in accordance with the requirements of MIL-S-19500. Semi-conductor devices listed in MIL-STD-701 shall be used whenever possible.

3.2.3 Electrical and electronic parts and electrical wire. In the selection of parts, fulfillment of major design objectives shall be the prime consideration in complying with the requirements specified in 3.2.3.1 through 3.2.3.8.

3.2.3.1 Micro-circuits. Micro-circuit technology shall be employed and micro-circuits shall conform to requirements specified herein and in MIL-STD-454. Micro-circuits shall be soldered or welded into the using circuits.

3.2.3.2 Micro-circuits screening. Micro-circuits shall be screened to the Class 'B' requirement of MIL-M-38510. Micro-circuits, procured without screening test data or with a certificate of compliance which does not specifically provide evidence of screening, shall be fully screened to Class 'B' requirements by the contractor or an approved alternative testing facility prior to installation in government equipment.

3.2.3.3 Hybrid micro-circuits. Hybrid micro-circuits shall meet MIL-M-38510, Class 'B' requirements.

3.2.3.4 Semiconductor devices. Semiconductur devices shall be JANTX or JANTXV types. When these are not available, the types used shall be conditioned, screened and tested equivalent to the TX requirements for similar JANTX types. All devices shall be hermetically sealed in glass, metal or ceramic. Plastic enclosed devices shall not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.2.3.5 Standard part requirements. Part requirements shall conform to MIL-E-5400. Standard parts shall be used where practical.

3.2.3.6 Non-repairable assemblies. When previously produced models of this equipment do not use non-repairable subassemblies, the design shall not be changed to employ non-repairable assemblies without the approval of the procuring activity. All non-repairable assemblies require approval.

3.2.3.7 Micro-circuits interchangeability. Micro-circuits of a specific type, procured from different sources, shall be interchangeable. This shall be demonstrated by the interchangeable use of these devices at all levels of equipment fabrication.

3.2.3.8 ER parts. Established reliability (ER) parts shall be used when available.

3.2.4 Maintenance module - A maintenance module is defined as the least complex segment of the contract and article(s) which is capable of being removed and replaced at organizational level maintenance using tools, support equipment and procedures approved for the end article(s). Further, it is the segment to which the contractor shall provide fault detection and isolation techniques consistent with scheduled and unscheduled organization level maintenance requirement.

3.3 Design and construction - The equipment shall conform with all the applicable requirements of MIL-M-5400 for design and construction, except as otherwise specified herein.

3.3.1 Total weight - The total weight of the equipment, excluding cables, shall be not greater than:

a. 30.0 pounds for Compensating Group, Magnetic AN/ASA-65(V)1.

b. 50.5 pounds for Compensating Group, Magnetic AN/ASA-65(V)2.

c. 33.5 pounds for Compensating Group, Magnetic AN/ASA-65(V)3 and AN/ASA-65A(V)3.

d. 56.5 pounds for Compensating Group, Magnetic AN/ASA-65(V)4.

e. 73.5 pounds for Compensating Group, Magnetic AN/ASA-65(V)5.

3.3.2 Reliability. The contractor shall conduct a reliability program in accordance with MIL-STD-785. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise specified in the contract. (see 6.2.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.3.2.1 Operational stability. The equipment shall operate with specified performance, continuously or intermittently for an accumulated operating period of at least 200 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating life. The equipment shall have a total operating life of 10,000 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement, as well as the replacement interval of these parts, shall be made available to the government inspector.

3.3.2.3 Reliability in mean time between failure (MTBF). Configurations a, b and c, shall have 1000 hours of specified mean (operating) time between failures when tested and accepted as specified in Table I.

3.3.2.4 Time totalizing meter. The units specified in Table I shall contain time totalizing meters in accordance with MIL-M-7793.

TABLE I. Time totalizing meter allocation

Unit	Unit Type Designation	Type of Meter
Control-Indicator	C-7718/ASA-65(V) C-8935/ASA-65(V) C-8936/ASA-65(V) C-8936A/ASA-65(V)	MS17322-6
Amplifier, Electronic Control	AM-6056/ASA-65(V) AM-6459/ASA-65(V) AM-6460/ASA-65(V)	MS17322-6
Computer, Magnetic Field	CP-1390/ASA-65(V)	MS17322-10

3.3.3 Cabling and connections

3.3.3.1 Cabling and connectors. The equipment shall be designed for the use of cables and connectors in accordance with WR-101 (Part I) and MIL-C-81511.

3.3.3.2 Interconnection cabling. The equipment shall be capable of specified operation using external wiring in accordance with the applicable requirements of WR-101 (Part I). The external wiring shall be unshielded, except that a minimum number of individual wires may be shielded when demonstrated as necessary to meet the interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.3.4 Control panels. All rack or console mounted control panels shall conform to the applicable requirements of MIL-C-6781, except that the panels shall be gray in accordance with FED-STD-595 (color number 36231) and shall have white lights.

3.3.5 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.3.6 Interference control. The equipment shall conform with all the requirements of WR-101 (Part I) for design and construction.

3.3.7 Maintainability

3.3.7.1 Maintainability program. The contractor shall establish and conduct a maintainability program in accordance with MIL-STD-471. All maintenance actions shall be capable of being performed by a technician with a comprehensive ability level as follows:

a. Civilian Education: High School graduate or equivalent.

b. Technical Training: Navy, Class A Technician School appropriate to the maintenance task

c. Experience: Three years of technical experience in addition to technical training.

3.3.7.2 Maintainability requirements

3.3.7.2.1 Mean time to repair (MTTR). The time to perform unscheduled maintenance actions shall be composed of:

a. Recognition of a fault
b. Isolation of the fault to a maintenance module
c. Maintenance module replacement
d. Retest for system readiness

Mean Maintenance Down Time ($\overline{M}_{CT}$) is the summation of all maintenance down times during a given period divided by the number of maintenance tasks performed during the same period. For Organizational Maintenance this time shall be less than 30 minutes. The maximum corrective maintenance down time ($M_{max_{CT}}$) at Organizational level shall be not greater than 60 minutes.

3.3.7.2.2 Packaging. The equipment shall be packaged in modular form so that, to the maximum extent practical, all sub-assemblies are of the plug-in type and are easily removable. If a special tool is required for removal, it shall be provided and attached to the equipment.

3.3.7.2.3 Module commonality. In order to ease the testing and spares requirements at all levels of maintenance, the number of different modules shall be kept to an absolute minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.3.7.2.4 Module accessibility. Module arrangements shall be such that access to each module does not require the removal of adjacent modules or parts other than access panels.

3.3.7.2.5 Keying. Modules and connectors shall be keyed to prevent the insertion of a connector or subassembly into an improper location within or on the equipment.

3.3.7.2.6 Part replacement. Part replacement on the modules shall be possible by technicians working in a designated service area such as an intermediate maintenance activity. Use of potting compounds, excessive moisture/fungus proofing compounds or any other substance to mechanically support piece parts is forbidden. Moisture/fungus proofing compounds, not in excess of a 30 mil thickness, are permitted provided reference designators and part identification remain clearly legible, part replacement is not impeded and compliance with MIL-E-5400 is maintained.

3.3.7.2.7 Access panels. Access panels shall be retained by quick disconnect fasteners, i.e., hand operated slide, dzus type inappropriate in section 3 where precise requirements are stated, or equivalent.

3.3.7.2.8 Service access. Devices requiring removal periodically for inspection and servicing shall be retained by quick disconnect fasteners, i.e., hand operated slide, dzus type, or equivalent.

3.3.7.2.9 Fault isolation. In conjunction with, but not necessarily as part of the Performance/Readiness testing features, there shall be features incorporated which will permit fault isolation to the module level. These features shall include the necessary switching arrangmenets with a GO-NO-GO readout device that will clearly indicate whether or not the function selected is within minimum performance standards. If a built-in readout device is not feasible, means shall be provided to use general or standard test equipment, i.e., Oscilloscope and/or multimeters.

3.3.7.2.9.1 Diagnostic ability. If feasible, the equipment shall be logically and electrically designed to facilitate the location of malfunctioning modules through diagnostic programming in conjunction with the aircraft's computer and other installed electronic equipment. Test points shall be made available at the front of the equipment to assist computer diagnosis of malfunctions.

3.3.7.2.10 Test points

3.3.7.2.10.1 Test point requirements. All test points shall be in accordance with MIL-STD-415 and shall accommodate standard government approved test equipment, i.e., multimeter or oscilloscope probes. In the event test points cannot accommodate standard probes, use of adapters will be permitted. These adapters shall be provided with and secured to the set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.3.7.2.10.2 Power supply test points. Accessible external individual test receptacles shall be provided on the equipment to measure the output voltages. The receptacles shall be identified as to the voltage normally present at the test point.

3.3.7.2.10.3 Chassis test points. Test points and built-in test facilities shall be included in the design of the equipment and shall be straightforward, logical, step-by-step troubleshooting sequence, as well as provide a single end-to-end performance check.

3.3.7.3 Maintainability assurance. Testing to demonstrate the achievement of the maintainability requirement specified herein shall be in accordance with 4.5.

3.3.8 Nomenclature, nameplates and identification marking. Nomenclature and serial number assignment, nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room Ambient (25 ±5°C)
b.	Altitude	Normal Ground
c.	Vibration	None
d.	Humidity	Room Ambient up to 90% Relative Humidity
e.	Input Power Voltage	115 ±1.0V AC 400 CPS ±1%

3.3.10 Service condition. The equipment shall operate as specified under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 1A equipment, except as modified herein.

3.3.10.1 Vibration. The equipment shall operate as specified when subjected to the vibration requirements of Curve I of MIL-E-5400 except for the Computer, Magnetic Field, which shall operate satisfactorily when subjected to vibrations with a double amplitude of 0.08 inch from 5 to 23 Hz and a 2 g level of vibration from 23 Hz to 500 Hz.

3.3.10.2 Altitude. The equipment shall operate as specified at all altitudes from sea level to 20,000 feet. Operating the equipment at any altitude up to 40,000 feet shall not cause permanent damage to the equipment; normal operation shall resume immediately after pressure equal to 20,000 feet altitude or lower is reached.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.3.11 Warm-up time. The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall be not greater than one minute under standard conditions and three minutes at extreme service conditions.

3.3.12 Primary inputs power requirements. The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from power sources with characteristics within the highest limits of voltage and frequency transients during normal electric system operation defined in MIL-STD-704. The equipment shall operate within the bounds of limits two and three of Fig. 3 of MIL-STD-704. Degraded operation may occur when the input voltage exceeds the above limits but remains within limits one and four of Figure 3 of MIL-STD-704. However, no damage shall be suffered by the equipment under transient conditions defined herein and normal operation shall automatically be resumed upon return of the input voltage to levels within limits two and three of Figure 3. The power required shall be not greater than the following specified amounts: (The use of 28V DC Aircraft power other than for annunciator lights, in this equipment, is prohibited.)

a. For the AN/ASA-65(V)1, (V)2, (V)3

(1) AC Power (Single Phase) 115V, Category B, 115 VA Maximum

(2) AC Power (Single Phase) 28V, Category B, 10 VA Maximum (for Panel Lamps only)

b. For the AN/ASA-65(V)4 and (V)5

(1) AC Power (Single Phase), 115V, Category B, 165 VA Maximum

(2) AC Power (Single Phase), 28V, Category B, 10 VA Maximum 1 for C-8935, C-8936 or C-8936A (Panel Lamps only)

(3) DC Power 2 to 28V DC, 5 Watt Maximum (for Annunciator Lights only)

(4) AC Power (Single Phase) 5V, Category B, 5 VA Maximum (for ID-2205 or ID-2254).

3.3.13 Protective devices. Circuit breakers shall be used in lieu of fuses whenever possible. Fuses, if used, shall be of approved types. Fuse holders and circuit breakers shall be accessible from the exterior of the equipment, plainly marked as to rating and size, and shall be of the fault indicating type and have access for quick voltage checks. Aircraft power for the ID-2205, ID-2254 and CP-1390 shall not be protected in the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.4 Performance. Unless otherwise specified herein, values set forth to establish the requirements for specified performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions shall be specified in the acquisition documents.

3.4.1 Operation. The equipment shall supply currents of the proper shape, polarity, phase and magnitude to obtain optimum compensation of the three permanent magnetic fields, and any six of the remaining induced and eddy-current magnetic fields of the aircraft in which MAD equipment is installed. Optimum compensation shall be defined as a reduction of the magnetic noise, generated by the aircraft's Perm, Induced and Eddy-current sources, during standard maneuvers, to a value equivalent to the straight and level noise level or ±0.01 gamma, whichever is greater. For the AN/ASA-65(V)1, (V)2, (V)3, and the AN/ASA-65A(V)3, the equipment shall determine the proper compensating currents by an automatic correlation process during standard compensation maneuvers. The actual maneuver time required to achieve the optimum compensation of an individual magnetic field component shall be not greater than 30 secs, except where the maneuver coefficient of the magnetic source is less than 0.05. For the AN/ASA-65(V)4 and (V)5, the Computer, Magnetic Field shall calculate the proper compensation currents, which are then displayed on the Indicator, Magnetic Field and manually transferred to the Control Indicator for entering into the coils. The computer shall gather data during a maneuver pattern consisting of five pitches of 5° peak-to-peak amplitude, five yaws of 5° peak-to-peak amplitude and five rolls of 10° peak-to-peak amplitude performed by a pilot or otherwise on each of the four cardinal or inter-cardinal magnetic headings. The time taken for each turn between successive headings shall be between 20 and 30 seconds. The total time to complete the above pattern shall not be greater than 10 minutes. Other maneuver patterns could be used. One or two iterations of the above procedure may be employed.

- Standard Maneuvers:

a. Roll. The aircraft shall be rolled ±10 degrees, with a period of 4 to 8 seconds (0.25 Hz to 0.125 Hz). Heading and pitch shall be maintained to within ±1 degree.

b. Pitch. The aircraft shall be pitched ±5 degrees with a period of 3 to 6 seconds. (0.33 Hz to 0.166 Hz). Altitude variations shall be held to a minimum, and in no case shall exceed 50 feet. Roll and Yaw shall be maintained to within ±1 degree.

c. Yaw. The aircraft shall be yawed through heading changes of ±5 degrees, with a period of 4 to 8 seconds (0.25 Hz to 0.125 Hz). A zero roll angle shall be maintained to within ±1 degree.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.4.2 Compensating field patching. The equipment shall use patch connectors for easy change of the combination of six induced and eddy-current terms to be compensated. The following terms shall be used:

a. AN/ASA-65(V)1 T, L, V, -LL, LV, VV, -tt, -ll, lv

b. AN/ASA-65(V)2 T, L, V, lv, -tt, -LL, -ll, vl, VV and AN/ASA-65(V)5

c. AN/ASA-65(V)3, AN/ASA-65A(V)3 T, L, V, lv, -tt, -LL, -ll, vl, LV and AN/ASA-65(V)4

3.4.3 Scale factors

3.4.3.1 Magnetometers. The three mutually perpendicular magnetometers and the related electronics shall produce an output voltage of 14 volts DC per Oersted which is linearly proportional to the magnetic input up to a field of 0.72 Oersted.

3.4.3.2 Compensating coils. The compensating coils shall produce the magnetic outputs along their individual longitudinal axis, at the separations stated, as specified in Table II. Currents in excess of those stated may result in loss of linearity.

TABLE II. Output coil data

Unit	Current	Separation	Output
MX-8130/ASA-65(V) L Coil T and V Coil	 100 mA 100 mA	 48 inches 48 inches	 50 gammas 100 gammas
MX-8897/ASA-65(V) Full Coil Between start and center tap	 76 mA 100 mA	 240 inches 240 inches	 76 gammas 50 gammas
MX-9057/ASA-65(V)	100 mA	192 inches	50 gammas
MX-9058/ASA-65(V)	72 mA	132 inches	40 gammas
MX-10079/ASA-65(V)	100 mA	205 inches	70 gammas

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.4.3.3 Complete equipment. The magnetic output of the equipment shall be a function of the term potentiometer settings. The scale factors, specified in Table III, expressed in number of digits of counter setting per unit of output at the MAD sensor shall be used.

TABLE III Scale factors

Equipment	Type of Field	Scale Factor
AN/ASA-65(V)1	Permanent fields Induced fields Eddy current fields except -11 -11 Eddy current field	10 digits per gamma 5 digits per gamma per Oersted 5 digits per gamma per Oersted per second 2.5 digits per gamma per Oersted per second
AN/ASA-65(V)2 and AN/ASA-65(V)5	Permanent fields Induced fields Eddy current fields except -11 -11 Eddy current field	10 digits per gamma 5 digits per gamma per Oersted 5 digits per gamma per Oersted per second 2.5 digits per gamma per Oersted per second
AN/ASA-65(V)3 AN/ASA-65A(V)3 and AN/ASA-65(V)4	Permanent fields Induced fields Eddy current fields	10 digits per gamma 5 digits per gamma per Oersted 5 digits per gamma per Oersted per second

The equipment shall have internal adjustments so that these scale factors can be maintained over the range of separation between output coils and MAD sensor, as specified in Table IV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

TABLE IV Separation range

Equipment	Coil	Range of Separation
AN/ASA-65(V)1	T, L, V	36 to 50 inches
AN/ASA-65(V)2 and AN/ASA-65(V)5	T, V	200 to 240 inches
	L	180 to 250 inches
AN/ASA-65(V)3 AN/ASA-65A(V)3 and AN/ASA-65(V)4	T, V	145 to 200 inches
	L	155 to 213 inches

3.4.4 DC offset. With all terms set at 500, the magnetic field produced at the detector shall be not greater than 0.1 gamma over the full temperature range.

3.4.5 Phase shift characteristic. In order to obtain proper correlation between the MAD sensor output signal and the aircraft maneuvers, the phase shift between the magnetic input into the magnetometers and the resulting signal at the input to the correlator shall be as specified in Table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

TABLE V Phase shift characteristics

Frequency	AN/ASA-65(V) Phase Shift	AN/ASA-65(V)2 AN/ASA-65(V)3 AN/ASA-65A(V)3 AN/ASA-65(V)4 AN/ASA-65(V)5 Phase Shift
0.125 Hz	52 ±10°	60 ±10°
0.15	42 ±10°	40 ±10°
0.175	34 ±10°	23 ±10°
0.2	26 ±10°	11 ±10°
0.225	20 ±10°	-05 ±10°
0.25	13 ±10°	-10 ±10°
0.275	7 ±10°	-19 ±10°
0.3	1 ±10°	-28 ±10°
0.325	-4 ±10°	-37 ±10°
0.35	-9 ±10°	-44 ±10°

NOTE: For the AN/ASA-65(V)4 and (V)5, this correlation is not used and therefore the phase shift characteristics are immaterial.

3.4.6 Magnetometer circuits. The requirements of 3.4.6.1 through 3.4.6.3 pertain to each magnetometer and its related electronics.

3.4.6.1 Sensitivity to beat frequency. If the magnetometer probes are subjected to a magnetic field in a direction parallel to their longitudinal axis with an amplitude of 4000 gammas and a frequency close to the input power frequency of 400 Hz, then the beat frequency amplitude as measured at the output of the magnetometer amplifier output shall be not greater than 10 mV peak-to-peak.

3.4.6.2 DC offset. The DC offset voltage at the output of the magnetometer amplifier shall be adjustable to zero. The variation from magnetometer to magnetometer, without readjustment, shall be not greater than 10 mV.

3.4.6.3 Rotational offset. When the magnetometer probe is rotated along its longitudinal axis in a magnetic field of 60,000 gammas perpendicular to that axis, then the output of the magnetometer amplifier shall not vary more than 60 mV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.4.7 Temperature compensation. The equipment shall be designed such that temperature variations over the range of environmental temperatures shall result in a variation of compensating fields at the MAD sensor of less than 0.025 gamma.

3.5 Detail requirements

3.5.1 Control-indicator. This unit shall be in accordance with MIL-C-6781 and the requirements specified in 3.5.1.1 through 3.5.1.7.

3.5.1.1 Function. The Control-Indicator, shall incorporate all the elements necessary to use the correlation result obtained from the Amplifier Electronic Control, to adjust the compensation until the performance objectives of the system are reached. The Control Indicator shall contain all the controls and indicators used in flight.

One of four types of functionally identical Control-Indicator Units shall be required depending on the specific application, namely:

a. C-7718/ASA-65(V) for use with Detecting Set, Magnetic AN/ASQ-10A aboard Type P-3C aircraft.

b. C-8935/ASA-65(V) for use with Detecting Set, Magnetic AN/ASQ-81(V) aboard Type P-3C aircraft.

c. C-8936/ASA-65(V) or C-8936A/ASA-65(V) for use with Detecting Set, Magnetic AN/ASQ-81(V) aboard Type S-3A aircraft.

3.5.1.2 Form factor. The unit shall be enclosed in a case whose dimensions shall not exceed 7 inches in length, 5.75 inches in width and 9 inches in height.

3.5.1.3 Weight. The weight of the unit shall be not greater than 9.5 pounds.

3.5.1.4 Mounting. The unit shall be suitable for hard mounting.

3.5.1.5 Contents. The Control-Indicator shall contain the following:

a. Electronic assembly. The Electronic Assembly portion of the Control-Indicator shall use the correlation result to control the input to the servo amplifier. The electronics shall be capable of performing compensation for the three permanent magnetic terms and any combination of six of the induced and eddy-current magnetic terms. This can be achieved by using plug-in modules or patch connectors which can be selected for a particular aircraft or installation.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the current version before use.

b. Potentiometer-servomotor assembly. Nine center-tapped potentiometers shall act as voltage dividers. The voltage on the wiper arm shall be representative of the compensating current. Three of the potentiometers shall be used for the three permanent magnetic terms. The remaining six potentiometers shall be used for any combination of the induced or eddy-current terms. All of the potentiometers shall be driven, singly, by the servomotor, and each potentiometer shall include an electromagnetic brake-clutch. The electromagnetic brake-clutches shall lock the position of the potentiometers, except when they are selectively actuated for automatic run-in or to be set to a predetermined value.

c. Counters. A total of nine three-digit counters shall indicate the setting of the potentiometers. 500 shall indicate the center of the potentiometer, 000 and 999 shall indicate the ends of travel of the potentiometers.

d. Time totalizing meter. A time totalizing meter shall be provided in accordance with MIL-M-7793. The meter shall time-monitor input power voltage application to the servo amplifier and servo motor, and need not be visible from the faceplate of the unit.

3.5.1.6 Controls. The following controls required for operation of the equipment shall be located on the front panel of the Control-Indicator and shall be identified as follows:

a. Mag term. A switch shall be provided to select the potentiometer to be adjusted, either manually or automatically. The switch shall control the current to the appropriate electro-magnetic clutch.

b. Rate. A switch shall be provided to permit the selection of four different automatic compensation speeds.

c. Up-Down. A toggle switch shall be provided to permit adjustment of the potentiometer selected by the Magnetic Term Selector.

d. Power-Off. A two-position toggle switch shall be provided to permit the control of AC input power to the equipment. Panel lighting power shall not be controlled by this switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

e. Servo-Off. A toggle switch shall be provided to Control the AC and internally generated DC Power to the servo amplifier and servo motor. The servo amplifier and servo motor will be activated only during compensation adjustment procedures.

f. + Off -. A polarity switch shall be provided to permit the proper selection of the correlation circuit polarity for automatic compensation.

3.5.1.7 Electrical connections. Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
1J1	M81511/03ED01P1	External Power for Panel Lighting, and interconnections to the Amplifier, Electronic Control.

3.5.2 Amplifier, electronic control. The unit shall meet the following requirements:

3.5.2.1 Function. The unit shall receive and amplify the signals from the Magnetometer Assembly. These signals shall be used for compensation of Induced and Eddy-current fields and shall be processed for use in a correlation with the MAD set output. This correlation result shall be supplied to the Control-Indicator, for the compensation adjustment. The unit shall contain the necessary circuits to change the voltage on the potentiometer wiper arms to a compensating signal to be fed into the appropriate Coil Assembly. The unit shall also contain the required power supplies to generate the DC voltages required for the operation of the equipment. One of three types of functionally identical Amplifier, Electronic Control units shall be required depending on the specific application, namely:

a. AM-6056/ASA-65(V), for use with Detecting Set, Magnetic AN/ASQ-10A aboard type P-3C aircraft.

b. AM-6459/ASA-65(V), for use with Detecting Set, Magnetic AN/ASQ-81(V) aboard type P-3C aircraft.

c. AM-6460/ASA-65(V), for use with Detecting Set, Magnetic AN/ASQ-81(V) aboard type S-3A aircraft.

3.5.2.2 Form factor. The unit shall be enclosed in a case whose dimensions shall be not greater than 17.25 inches in length, 5 inches in width (not including flanges) and 8.0 inches in height.

3.5.2.3 Weight. The weight of this unit shall be not greater than 16.5 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.5.2.4 Mounting. The unit shall be capable of being hard mounted.

3.5.2.5 Contents. The Amplifier, Electronic Control unit shall contain the following:

a. Time totalizing meter. A time totalizing meter in accordance with MIL-M-7793 shall be provided. The meter shall time-monitor the input voltage application to the Amplifier, Electronic Control unit. The Meter shall be mounted so as to be visible from outside of the unit.

b. Electronics. The Amplifier, Electronic Control shall receive and amplify the signals from the Magnetometer Assembly. The unit shall also process the MAD set output signal in a filter whose phase characteristics are such that optimum phase matching is achieved between the input signals to the correlator. The unit shall then supply the correlation result to the Control-Indicator for the correlation adjustment. The Electronic Assembly also contains the circuitry to change the voltage on the potentiometer wiper arms to a compensating signal to be fed into the appropriate coil assembly.

c. Power supplies. These shall generate the DC voltage required for the operation of the equipment.

3.5.2.6 Controls. No external controls shall be provided on this unit.

3.5.2.7 Electrical connections. Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
a) 2J1	M81511/03EE01P2	External AC power, and Interconnections to the Control-Indicator
b) 2J2	M81511/03EE01S1	Interconnections to the MAD set, the Magnetometer Assembly, and the Coils.

3.5.3 Magnetometer assembly DT-355/ASA-65(V). This unit shall meet the requirements specified in 3.5.3.1 through 3.5.3.7.

3.5.3.1 Function. The unit shall be a tri-axial magnetometer system. The unit shall consist of three mutually perpendicular magnetometers, each capable of reading its vector component of the ambient magnetic field to within ±0.5 percent. The unit shall provide the signals necessary for the correlation processes carried out in the Amplifier Electronic Control, Unit #2. Units shall be interchangeable without the need for readjustment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.5.3.2 Form factor. The individual magnetometers shall be not greater than 0.75 inches in diameter and 6 inches in length. The unit shall be constructed to permit separate mounting of the individual magnetometers.

3.5.3.3 Weight. The Weight of the unit shall be not greater than two pounds, including the alignment and mounting hardware.

3.5.3.4 Controls. No controls shall be provided with this unit.

3.5.3.5 Electrical connections. Connections from the magnetometer assembly to the Amplifier Electronic Control shall be provided through a connector type M81511/06ED01P1.

3.5.3.6 Accuracy of alignment. The individual magnetometers shall be mounted mutually perpendicular to within one degree of arc. The mounting holes and surface of the magnetometer assembly shall be located precisely enough to be used as references for the aircraft installation.

3.5.3.7 External markings. The mounting directions shall be clearly marked on the outside of the individual magnetometers.

3.5.4 Coil assembly, MX-8130/ASA-65(V). This unit shall meet the requirements specified in 3.5.4.1 through 3.5.4.6.

3.5.4.1 Function. One assembly shall be required for operation with Detecting Set, Magnetometer AN/ASQ-10A. This unit shall provide for the utilization of the output of the Amplifier, Electronic Control, to produce transverse, longitudinal and vertical magnetic fields which are used to cancel the aircraft's permanent, induced and eddy-current magnetic fields. The unit shall contain three coils.

3.5.4.2 Form factor. The overall dimensions of this unit shall be not greater than a 3.5-inch cube, exclusive of mounting provisions.

3.5.4.3 Weight. The weight of the unit shall be not greater than two pounds.

3.5.4.4 Controls. No controls shall be provided on this unit.

3.5.4.5 Electrical connections. Connections to the Amplifier Electronic Control shall be provided through a connector type MS3101F14S-6P.

3.5.4.6 External markings. The mounting direction shall be clearly marked on each assembly.

3.5.5 Coil, magnetic compensating, MX-8897/ASA-65(V). This unit shall meet the requirements specified in 3.5.5.1 through 3.5.5.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.5.5.1 Function. Three identical units shall be required for operation with Detecting Set, Magnetic AN/ASQ-81(V) aboard Type P-3C aircraft. Each unit shall provide for the utilization of the output of the Amplifier, Electronic Control, to produce transverse, longitudinal, or vertical magnetic fields which are used to cancel the aircraft's permanent, induced, and eddy-current magnetic fields.

3.5.5.2 Form factor. The overall dimensions of this unit shall be not greater than a 1.765 inches in diameter and 25.06 inches in length, exclusive of electrical connections and mounting provisions.

3.5.5.3 Weight. The weight of each unit shall be not greater than 7.5 pounds.

3.5.5.4 Mounting. The three units used in the equipment shall be hard mounted with the three coils mutually perpendicular to within 1° of arc.

3.5.5.5 Controls. No controls shall be provided on this unit.

3.5.5.6 Electrical connections. Connections to the Amplifier, Electronic Control, shall be provided through a connector type MS3106F14S-6P.

3.5.5.7 External markings. The mounting direction shall be clearly marked on each unit.

3.5.6 Coil, magnetic compensating, longitudinal, MX-9057/ASA-65(V). This unit shall meet the requirements specified in 3.5.6.1 through 3.5.6.7.

3.5.6.1 Function. One unit shall be required for operation with Detecting Set, Magnetic AN/ASQ-81(V) aboard Type S-3A aircraft. This unit shall provide for the utilization of the output of the Amplifier, Electronic Control, to produce a longitudinal magnetic field which is used to cancel the aircraft's longitudinal component of permanent, induced and eddy-current magnetic fields.

3.5.6.2 Form factor. The overall dimensions of this unit shall be not greater than a 18.03 inches in length and 1.015 inches in diameter, exclusive of electrical connections and mounting.

3.5.6.3 Weight. The weight of this unit shall be not greater than 2.5 pounds.

3.5.6.4 Mounting. This unit shall be hard mounted with the coil perpendicular to the two compensating coils, MX-9058/ASA-65(V), to within one degree of arc.

3.5.6.5 Controls. No controls shall be provided on this unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.5.6.6 Electrical connections. Connections to the Amplifier, Electronic Control shall be provided through a connector type LS13585R15-6P.

3.5.6.7 External markings. The mounting direction shall be clearly marked on each unit.

3.5.7 Coil, magnetic compensating, transverse, vertical, MX-9058/ASA-65(V). This unit shall meet the requirements specified in 3.5.7.1 through 3.5.7.7.

3.5.7.1 Function. Two units shall be required for operation with Detecting Set, Magnetic AN/ASQ-81(V) aboard type S-3A aircraft. Each unit shall provide for the utilization of the output of the Amplifier, Electronic Control, to produce transverse, or vertical magnetic fields which are used to cancel the aircraft's transverse or vertical components of permanent, induced and eddy-current magnetic fields.

3.5.7.2 Form factor. The overall dimensions of this unit shall be not greater than a 1.015 inches in diameter and 9.5 inches in length, exclusive of electrical connections and mounting.

3.5.7.3 Weight. The weight of the unit shall be not greater than 1.5 pounds.

3.5.7.4 Mounting. The two units used in the equipment shall be hard mounted with their coils and the compensating coil MX-9057/ASA-65(V), all mutually perpendicular to within one degree of arc.

3.5.7.5 Controls. No controls shall be provided on this unit.

3.5.7.6 Electrical connections. Connections to the Amplifier, Electronic Control, shall be provided through a connector type LS13585R15-6P.

3.5.7.7 External markings. The mounting direction shall be clearly marked on each unit.

3.5.8 Coil, magnetic compensating, longitudinal, MX-10079/ASA-65(V). This unit shall meet the requirements specified in 3.5.8.1 through 3.5.8.7.

3.5.8.1 Function. One unit shall be required for operation with Detecting Set, Magnetic AN/ASQ-81(V) aboard type S-3A aircraft. This unit shall provide for the utilization of the output of the Amplifier, Electronic Control, to produce a longitudinal magnetic field which is used to cancel the aircraft's and Torpedo's longitudinal component of permanent, induced and Eddy-current magnetic fields. This unit is used in lieu of the MX-9057/ ASA-65(V), to also provide for Torpedo compensation ability.

3.5.8.2 Form factor. The overall dimensions of this unit shall be not greater than a 24.03 inches in length and 1.015 inches in diameter, exclusive of electrical connections and mounting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.5.8.3 Weight. The weight of the unit shall be not greater than 3.5 pounds.

3.5.8.4 Mounting. This unit shall be hard mounted in the boom.

3.5.8.5 Controls. No controls shall be provided on this unit.

3.5.8.6 Electrical connections. Connections to the Amplifier, Electronic Control, shall be provided through a connector type LS13585R15-6P.

3.5.8.7 External markings. The mounting direction shall be clearly marked on each unit.

3.5.9 Computer, magnetic field, CP-1390/ASA-65(V). This unit shall meet the requirements specified in 3.5.9.1 through 3.5.9.7.

3.5.9.1 Function. The unit shall receive the MAD signal from the AN/ASQ-81(V), in the form of a Larmor Frequency. It shall also receive anlog signals from the Amplifier, Electronic Control representing the value of the Earth's Magnetic Field Components along the aircraft's longitudinal, vertical and transverse axis. It shall use all these signals to calculate the update value of the terms, necessary to compensate the aircraft maneuver related interference to a minimum. It shall have built-in test equipment to monitor the proper operation of the AN/ASA-65(V)4 or (V)5. In addition, it shall contain the necessary Power Supplies and Memory to achieve the above.

3.5.9.2 Form factor. The unit shall be enclosed in a case whose dimensions shall be not greater than 9 inches in height, 9 inches in width and 16.5 inches in depth, including all protrusions.

3.5.9.3 Weight. The weight of this unit shall be not greater than 19 pounds.

3.5.9.4 Mounting. This unit shall be capable of being hard-mounted on a mounting base.

3.5.9.5 Contents. The unit shall contain the following:

a. Time Totalizing Meter. A Time Totalizing Meter in accordance with MIL-M-7793 shall be provided to monitor the application of input power to this unit. It shall be mounted so as to be visible from outside the unit.

b. Electronics. The unit shall contain all electronic circuits necessary to fulfill its intended use. These circuits shall include the following:

(1) A frequency to Analog Converter to convert the AN/ASQ-81(V) Larmor Frequency to an analog signal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

(2) An Analog to Digital Converter to translate all analog signals, necessary for Data Gathering and BITE, into digital signals.

(3) A microprocessor for data gathering and calculations and for BITE monitoring.

(4) Memory in the form of Programmable Read Only Memory to contain the application program and the micro code, and Random Access Memory for temporary storage.

(5) Power Supplies to generate the DC Voltages necessary for the operation of the equipment.

3.5.9.6 Controls. No external controls shall be provided on this unit.

3.5.9.7 Electrical connections. Electrical connections shall be provided as specified in Table VI.

TABLE VI Electrical connections

Reference Designation	Receptacle Type	Function
5J1	M81511/53WF01P1	Interconnections to the Amplifier, Electronic Control, the Control-Indicator, the Magnetic Assembly and the Coils.
5J2	M81511/53WF01P2	Interconnections to the Indicator, Magnetic Field
5J3	M39012/34-0001	Interconnection to the AN/ASQ-81(V) Larmor Signal
5J4	M39012/24-0001	Test Connector for connection to an oscilloscope.

3.5.10 Indicator, magnetic field. This unit shall meet the requirements of MIL-C-6781 and the following requirements:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

3.5.10.1 Function. The Indicator, Magnetic Field shall incorporate all the elements necessary to display, one at a time, the changes in the nine compensation terms necessary to obtain optimum compensation. These indicated changes are then to be transferred to the Control Indicator. The unit shall also contain all controls needed in flight, to operate the Computer Magnetic Field. One of two nearly identical Magnetic Field Indicators shall be required, depending on the application:

a. ID-2055/ASA-65(V) for use aboard the S-3A aircraft.

b. ID-2254/ASA-65(V) for use aboard the P-3C aircraft.

3.5.10.2 Form factor. The unit shall be enclosed in a case whose dimensions shall be not greater than 5.75 inches in length and 4.27 inches in width, and 3.75 inches in height, including the external connector.

3.5.10.3 Weight. The weight of the unit shall be not greater than two pounds.

3.5.10.4 Mounting. This unit shall meet the mounting requirements of MIL-C-6781.

3.5.10.5 Contents. The unit shall contain the following:

Display Assembly. The display assembly shall contain a four digit and sign light emitting diode numeric display. Each digit shall contain seven segments and shall be 0.27 inches in height.

3.5.10.6 Controls. The following controls required for operation of the equipment shall be located on the front panel of the Magnetic Field Indicator and shall be identified as follows:

a. PWR-OFF. A two-position toggle switch shall be provided to permit control of AC input power to the equipment. Panel lighting power shall not be controlled by the switch.

b. WPN LOAD. On the ID-2205/ASA-65(V) a five position rotary switch shall be provided, the positions marked '0', '1', '2', '3' and '4' to provide for coarse compensation of none, one, two, three or four, mark 46 torpedos respectively.

On the ID-2254/ASA-65(V) a nine position rotary switch shall be provided, the positions marked '0', '1', '2', '3', '4', '5', '6', '7' and '8' to provide for coarse compensation of up to eight mark 46 torpedos.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

c. MODE. A fourteen-position rotary switch shall be provided to enable the following operation or modes:

(1) OFF. All modes and terms are disabled and the Display and EXECUTE lights are extinguished. The FAULT light is still active and will light up if a "hardware" error occurs.

(2) BITE. To perform "On Demand" system test, using built-in test equipment. When this mode is selected the EXEC switch shall light up green. Pressing the EXEC switch shall start the bite program. When the program is running, the EXEC shall light up yellow. At completion, the EXEC light shall extinguish and the display shall indicate "-8888" for no error or shall indicate an error code.

(3) COMP. To gather data for calculation of the nine terms and to calculate these terms. On selecting the COMP mode the EXEC shall light up green. Pressing the EXEC switch shall start the data gathering. When the data gathering is in progress, the EXEC shall light up yellow. Data gathering is terminated, either by pressing the EXEC switch again or automatically after ten minutes. At termination the EXEC light will stay yellow and the computer will calculate the value of the adjustment of the nine terms. At completion of this calculation, the EXEC switch light shall extinguish.

(4) WD. This weapon drop mode is identical to the COMP mode with the exception that the computer will calculate the value of four adjustments only, under the assumption that the other five (5) terms have not changed.

(5) T, L, V, 1, 2, 3, 4, 5, 6. In these nine positions the value of the adjustment of the respective term is displayed. When a term is selected, the EXEC switch shall light up green. After pressing the EXEC switch again, it will extinguish and the term adjustment is shown on the display.

(6) CAL. In this calibration mode, the EXEC switch shall light up green and the display shall start counting from 0 through 33 in one second intervals and repeat this again. Pressing the EXEC switch, when a particular channel number appears, will stop the counter, display the voltage of the selected channel and change the EXEC from green to yellow. On pressing the EXEC again it will return to green and the display will resume counting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

d. FAULT. A yellow system warning light shall be provided to indicate a failure. It shall be in a steady mode to indicate a system failure and in a flashing mode to indicate an operational failure.

e. EXEC. An illuminated pushbutton switch shall be provided to initiate the execution of the following modes:

BITE, COMP, WD, T, L, V, L, 2, 3, 4, 5, 6 and CAL.

The switch shall be illuminated green when available; Yellow or Amber when in use, and not illuminated when unavailable.

3.5.10.7 Electrical connections. Electrical connections shall be provided as follows:

Reference Designation	Receptacle Type	Function
6J1	M81511/53WF01P3	Connections to the Computer, Magnetic Field and Aircraft 28V and 115V power.

3.5.11 Mounting base, MT-6086/ASA-65(V). This unit shall adapt the Computer Magnetic Field for quick removal and installation in the S-3A aircraft by means of self-locking front retainers and spring-loaded rear retainers.

3.5.11.1 Form factor. The unit shall be not greater than 17.5 inches in length, 9.5 inches in width, and 1.8 inches in height.

3.5.11.2 Weight. The weight of the unit shall be not greater than two pounds.

3.5.12 Mounting base, MT-6174/ASA-65(V). This unit shall adapt the Computer Magnetic Field for quick removal and installation in the P-3C aircraft by means of self-locking front retainers and spring-loaded rear retainers.

3.5.12.1 Form factor. The unit shall be not greater than 16 inches in length, 9.5 inches in width, and 1.8 inches in height.

3.5.12.2 Weight. The weight of the unit shall be not greater than two pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Reponsibility for inspection. Unless otherwise specified in the contract or the purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Initial Production Test (see 4.3)
c. Acceptance Tests (see 4.4).

4.2 First article inspection. First article inspection tests shall be made on an equipment representative of the production equipments to be supplied under the contract. First article inspection shall be accomplished by the contractor and shall be conducted in accordance with the approved test procedures of 4.6.

4.2.1 First article inspection data. All required test data collected during these tests shall be made available to the procuring activity (see 6.2.2). This data shall include, but not be limited to, a list of all electrical and electronic parts giving:

a. Specified voltage, current and temperature ratings.
b. Applied voltage and current.
c. Ambient and surface temperature of temperature-critical parts. This data shall be obtained under the extreme high temperature operating conditions.

4.2.2 Scope of tests. First article inspection shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. First article inspection shall include the following:

a. Environmental Test in accordance with MIL-T-5422.

b. Interference Tests in accordance with MIL-STD-461 and MIL-STD-462.

c. Shakedown Test

d. Power Input Test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.2.2.1 Shakedown test. Each preproduction equipment shall be subjected to a shakedown test of 100 continuous hours of failure-free operation. The test shall be conducted in accordance with MIL-STD-781, using test level E and 5.2.3.2 therein. Failures encountered in the test shall be the subject of detailed failure analysis and corrective action. The corrective action loop shall document the assignable cause of failure and the action taken. The test shall then be started from time zero. The performance characteristics to be measured shall consist of a series of functional tests exercising all portions of the equipment in a manner as stringent as that required during its ultimate application.

4.2.2.2 Power input test. Each equipment shall be subjected to a range of input voltage and frequencies as follows:

a. The input voltage shall be varied within the limits of Figure 3 of MIL-STD-704 as defined in 3.3.12 herein. The input frequency shall be varied between limits two and three of Figure 5 of MIL-STD-704. The equipment shall perform as specified in 3.4.

b. It shall be demonstrated that input voltage transients greater than those specified above, but remain in their limits one and four of Figure 3 of MIL-STD-704, shall not cause damage to the equipment and when the voltage returns within the specified limits, normal operation shall automatically resume (see 3.3.12).

4.2.3 First article approval. Approval of the First Article sample shall be by the procuring activity upon satisfactory completion of all tests. (See 4.2.2.)

4.2.4 Production equipments. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equal to the approved first article sample. Each equipment shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the procuring activity.

4.3 Initial production tests. One of the first ten production equipments shall be selected and sent at the contractor's expense, to a designated Government laboratory for tests (see 6.2.1). This equipment will be selected by the procuring activity after the equipment has successfully passed all individual tests. The first article sample shall not be selected for this test.

4.3.1 Scope of tests. This equipment shall be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equal to the previously approved first article sample and that it meets all applicable requirements, including design, construction, workmanship, performance and quality.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.3.2 Accessory material. In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and include data necessary to test the equipment.

4.3.3 Initial production sample approval. Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made eveident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for the deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted, when requested by the procuring activity.

4.4 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance test. Contractors not having testing facilities satisfactory to the procuring activity shall engage the services of a commercial testing laboratory acceptable to the procuring activity. Such reports shall be signed by an authorized representative of the contractor or contractor and laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests (see 4.4.1)

b. Sampling Tests (see 4.4.2)

c. Reliability Assurance Tests (see 4.4.3)

d. Special Test (see 4.4.4)

4.4.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product (see 4.4.1.1)

b. Operational Test (see 4.4.1.2)

c. Manufacturing Run-In Test (see 4.4.1.3)

4.4.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.4.1.2 Operational test. Each equipment shall be operated long enough to permit the equiment temperature to stabilize and to check sufficient characteristics and record adequate data to assure specified equipment operation.

4.4.1.3 Manufacturing run-in test. Each equipment shall be operated under the conditions specified below for a period of six hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least six hours of operation.

a.	Temperature	Ambient Room
b.	Humidity	Ambient Room
c.	Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points and a minimum amplitude of + 3 g's)

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the six-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the six-hour period of operation following the ten-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. The input power voltage and frequency shall also be cycled periodically between the specified tolerances. The power input test specified in 4.2.2.2 shall be conducted. Should a failure occur, the equipment shall be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetition failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The six-hour period specified above may be composed of two three-hour periods to conform with standard working hours.

4.4.2 Sampling tests. Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector as specified in Table VII. (Perform Sampling Tests only if Reliability Sampling Tests of 4.4.3 are deleted by contractual action.) (See 6.2.1.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

TABLE VII Sampling test selection

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and shall be based on results of the preproduction, initial production, individual and special tests.

c. Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability assurance tests. Reliability Assurance Tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.4.3.1 Qualification phase. Prior to the acceptance of equipment under the contract or order, a minimum of three equipments shall be tested as specified in MIL-STD-781, under the section entitled 'Qualification Phase of Production Reliability Tests'. The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level E of MIL-STD-781 shall be used with temperature cycling in accordance with Figure I of MIL-STD-781. The Accept-Reject Criteria for Test Plan IV of MIL-STD-781 shall be used.

4.4.3.2 Reliability production acceptance phase tests. The equipment throughout production, shall be tested as specified in MIL-STD-781 as modified herein under the section entitled 'Production Acceptance (Sampling) Phase of Production Reliability Tests', except that all equipment produced shall be tested. Test level E of MIL-STD-781 shall be used with temperature cycling in accordance with Figure I of MIL-STD-781.

4.4.3.2.1 All equipment test. Each equiment produced shall be tested for 100 hours. Prior to the 100 hours test on each equipment, a burn-in period may be used. If used, the burn-in shall be the same for each equipment, in time and severity. To determine whether the MTBF is being met at any time during the contract, the operation test hours and the failures thereon (not counting burn-in failures or burn-in operation time) shall be totaled and the results compared with the reject line of Test Plan V of MIL-STD-781. (Extend the line as necessry to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month, the test results shall be made available to the procuring activity (see 6.2.2). At any time that the current totals of test hours and test failures plotted on Test Plan V curves show a reject situation, the procuring activity shall be notified. The procuring activity may stop the acceptance of equipment any time that a reject situation exists, pending a review of the contractor's efforts to improve the equiment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria preventative maintenance to be allowed during the test, etc., shall be part of the test procedures, prior to the beginning of the Modification Test Phase of the Reliability Assurance Tests.

4.4.4 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design of material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipment previously subjected to the sampling or reliability assurance tests.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:35Z
Check the source to verify that this is the current version before use.

4.4.4.1 Special test schedule. Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Shakedown tests. Each of the production equipments submitted for acceptance testing prior to approval of the Reliability Assurance Tests of 4.4.3 shall be subjected to the shakedown tests specified in 4.2.2.1. Upon satisfactory completion of the Reliability Assurance Tests the shakedown tests shall be discontinued.

4.4.4.3 Scope of tests. Special tests shall consist of such tests as specified in the contract (see 6.2.1). Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure, for procuring activity review prior to conducting the tests.

4.4.5 Equipment failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Where practical, include a test in the individual test to check all equipment for these requirements until assurance is obtained that the defect has been corrected.

4.5 Maintainability demonstration. A Maintainability Demonstration shall be conducted in accordance with Method I, Test Plan (A1) plus (B1) of Table II of MIL-STD-471 and as follows, to verify that requirements of 3.3.7.2.1 are satisfied.

4.5.1 Test conditions. Before beginning the test, the equipment shall be checked to determine that it is operating as specified. All dust covers shall be in place on the equipment prior to the initiation of this test. Test equipment, tools, spare parts, diagnostic programs, and documentation shall be available for the test.

4.5.2 Maintenance actions. A maintenance action shall be as follows:

a. Fault Isolation

b. Repair

c. Equipment Checkout

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

4.5.3 Simulated failures. The failures shall be randomly introduced, one at a time, throughout the equipment.

4.5.3.1 Inducing failures. Single failures shall be induced into the equipment by substituting modules that are known to be defective or, where defective modules are not available, simulated failures shall be induced by applying grounds at test jacks or on extended modules. Any failure induced for demonstration purposes shall consist of a continuous malfunction.

4.5.4 Maintenance action dowtime measurements. The time required for the technician to perform each maintenance action, including the use of technical manuals and support equipment, shall be measured starting at the time the technician is made aware that a malfunction exists and ending at the time the equipment is restored to normal operation. Any delay such as waiting for parts, obtaining test equipment, or administrative delays shall be recorded, but excluded from the downtime computation.

4.6 Test procedures. The procedure used for conducting all tests requied by this specification shall be prepared by the contractor (see 6.2.2). MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity (see 6.2.1).

4.7 Presubmission testing. No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all requirements.

4.8 Test data reporting. MIL-STD-831 shall be used as a guide in the preparation of test reports.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment in accordance with MIL-E-17555 and as specified in the contract or purchase order (see 6.2.1).

6. NOTES

6.1 Intended use. The equipment covered in this specification is intended for use in aircraft incorporating magnetic anomaly detection systems to compensate for fixed magnetic fields of the airframe and for magnetic fields caused by maneuvers. The compensation will improve the system effectiveness of the magnetic anomaly system.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Site of testing facilities (see 4.3).

c. Selection of applicable levels of packaging and packing (see 5.1).

d. Configuration required, e.g., (V)1, (V)2, (V)3, A(V)3, (V)4 or (V)5.

e. Approval for use of test procedures from previous contracts (see 4.6).

f. Reliability if other than specified in 3.3.2.

g. If Reliability Sampling tests are to be deleted (see 4.4.2).

h. Special tests: if required (see 4.4.4.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-140.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data requirement title	Applic. DID No.	option
a. 3.3.6	E.M.I. Control Plan	DI-R-7061	-
b. 4.2.1	Report, First Article (Preproduction) Test	UDI-T-21349	-
c. 4.4.3.2.1	Reports, Test	DI-T-2072	-
d. 4.4.3.3	Report, Operability Test	UDI-R-20457	-
e. 4.4.4.3	Qualification and Acceptance Test Procedures	DI-T-3301	-
g. 4.5	Maintainability Demonstration Plan	DI-R-5318	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

6.2.3 First article. When a first article inspection is required, the item will be tested and should be a sample selected from the first ten production items. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

6.3 Precedence of documents. When the requirements of the contract, this specification, or application subsidiary specifications are in conflict, the following precedence should apply:

a. Contract. The contract should have precedence over any specification.

b. This specification. This specification should have precedence over all applicable subsidiary specification. Any deviation from this specification, or from subsidiary specifications where applicable should be specifically approved in writing by the procuring activity.

c. Referenced specifications. Any referenced specification should have precedence over all applicable subsidiary specifications referenced therein. All referenced specification should apply to the extent specified.

6.4 Performance objectives. Minimize size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which should be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.5 Non-Repairable subassemblies. As a general rule, non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts, tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-To-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.

6.6 Type designations. The type designations may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Associated equipment. The equipment should operate with the following associated equipment:

a. RO-32/ASQ, AN/ASQ-8, AN/ASQ-10, AN/ASQ-10A,

b. AN/ASQ-81(V), AN/ASA-64A and AN/ASQ-501(V)

6.9 Performance characteristics to be measured. Performance characteristics to be measured, to insure that the equipment complies with all requirements of 3.5 (Detail Requirements) and all operational requirements, should be submitted to the procuring activity for approval.

6.10 Test values. Normal and limiting values of performance data should be determined at the standard conditions specified in 3.3.9. This data is to be used in testing the equipment at installation points for compliance with minimum acceptance standards of performance.

6.11 Supersession data. The revision supersedes the basic document MIL-C-81461(AS) since Revisions A, B, C, and D were never issued and entered into the documentation system.

Preparing activity

Navy - AS

(Project 5845-N080)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:35Z
Check the source to verify that this is the current version before use.